Begin

In[1]:=
```
Print[
 "CopyRight (C) 2022, Patrick L. Nash, under the General Public License."]
```

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In[2]:=
```
Print["Please  cite  this  work,  and this web page, if you use it"]
```

Please cite this work, and this web page, if you use it

defs; some Symbols; metric

In[3]:=
```
Needs["Notation`"]
```

ψ_{nj0ss3}

In[4]:= Symbolize[ψ_{nj0ss3}]

In[5]:= Symbolize[u^a]

In[6]:= Symbolize[J_y]

$$g^{\alpha\beta} = g_{\alpha\beta} = \mathrm{diag}(1,1,1,-1) \tag{5}$$

In[7]:= Symbolize[$g_{\alpha\beta}$]

note below : g_{AB} is really $g_{(A)\,(B)}$ :: Minkowski - Lorentz metric

In[8]:= Symbolize[$S44\alpha\beta$]

$$\Sigma^{\alpha\beta} = \xi S^{\alpha\beta} u,$$

In[9]:= Symbolize[$\Sigma^{\alpha\beta}$]

$$\tfrac{1}{2}\bar{\Sigma}^{jk}\Sigma_{jk} = \tfrac{1}{4}\,\partial_a\, u^a u^b\, \partial_b - \tfrac{1}{4}\partial_a\, u^b u^b\, \partial_a,$$

In[10]:= Symbolize[Σ^2]

4×4 Dirac γ matrices involve these symbols :

$$S^{\alpha\beta} = -\tfrac{1}{4}[\gamma^{\alpha},\gamma^{\beta}], \tag{22}$$

$$S^{\alpha\beta} = -\tfrac{1}{2}\gamma^{\alpha\beta}. \tag{23}$$

The symplectic form ϵ on D_4 is defined to be

$$\epsilon = \gamma^{\delta 4}. \tag{24}$$

As a consequence of Eq. (1), and the definitions of Eqs. (16)–(24), are the identities

$$\tilde{\gamma}^{\alpha}\epsilon = -\epsilon\gamma^{\alpha}, \tag{25}$$

$$\tilde{S}^{\alpha\beta}\epsilon = -\epsilon S^{\alpha\beta}, \tag{26}$$

$$[S^{\alpha\beta},\gamma_{\mu}] = \delta^{\alpha}_{\mu}\gamma^{\beta} - \delta^{\beta}_{\mu}\gamma^{\alpha}, \tag{27}$$

$$[S^{\alpha\beta},S^{\mu\nu}] = g^{\alpha\mu}S^{\beta\nu} - g^{\alpha\nu}S^{\beta\mu} - g^{\beta\mu}S^{\alpha\nu} + g^{\beta\nu}S^{\alpha\mu}, \tag{28}$$

and

$$\gamma^{5}S^{\alpha\beta} = \tfrac{1}{2}g^{\alpha\mu}g^{\beta\nu}\epsilon_{\mu\nu\lambda\sigma}S^{\lambda\sigma}. \tag{29}$$

$$g^{\alpha\beta} = g_{\alpha\beta} = \mathrm{diag}(1,1,1,-1) \tag{5}$$

In[11]:=
```
(*Symbolize[ gαβ ]*)
```

SO (4, 4) Minkowski Lorentz metric η4488 : 4 + 4 spacetime; 8 × 8 dimensional :

In[12]:=
```
(η4488 = ArrayFlatten[
    {{IdentityMatrix[4], 0}, {0, -IdentityMatrix[4]}}]) // MatrixForm
```

Out[12]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

In[13]:=
```
η4488;
Dimensions[%];
(gαβ = η4488[[2 ;; 5, 2 ;; 5]]) // MatrixForm
```

Out[15]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

In[16]:=
```
Protect[gαβ];
```

In[17]:=

```
Zero4 = ConstantArray[0, {4, 4}]; ID4 = IdentityMatrix[4];
ϵ3 = Array[Signature[{##}] &, {3, 3, 3}]
ϵ4 = Array[Signature[{##}] &, {4, 4, 4, 4}];
```

Out[18]=

```
{{{0, 0, 0}, {0, 0, 1}, {0, -1, 0}},
 {{0, 0, -1}, {0, 0, 0}, {1, 0, 0}}, {{0, 1, 0}, {-1, 0, 0}, {0, 0, 0}}}
```

SO(4)

In[20]:=

```
Qa[h_, p_, q_] := Signature[{h, p, q, 4}];
Qb[h_, p_, q_] := ID4[[p, 4]] * ID4[[q, h]] - ID4[[p, h]] * ID4[[q, 4]];
SelfDualAntiSymmetric[h_, p_, q_] := Qa[h, p, q] - Qb[h, p, q] ;
AntiSelfDualAntiSymmetric[h_, p_, q_] := (Qa[h, p, q] + Qb[h, p, q] );
```

In[24]:=

```
Qa[1, 2, 3]
```

Out[24]=

```
1
```

In[25]:=

```
ϵ4[[1, 2, 3, 4]]
```

Out[25]=

```
1
```

Under the involutive automorphism $\gamma^{AB} \rightarrow -\tilde{\gamma}^{AB}$ of so(3,3), the Lie algebra decomposes into the eigenvalue (-1) and eigenvalue $(+1)$ subspaces corresponding to, respectively, the nine linearly independent real traceless symmetric 4×4 matrices, and the six linearly independent real skew-symmetric 4×4 matrices. The eigenvalue $(+1)$ subspace is the subalgebra so(4), which is the Lie algebra of $\overline{\mathrm{SO}(4)}$, the maximal compact subgroup of $\overline{\mathrm{SL}(4,\mathbf{R})}$. The subalgebra so(4) $\cong$ su(2) $\dot{+}$ su(2) may be further decomposed into the even (eigenvalue $+1$) and odd (eigenvalue -1) subspaces of the linear transformation of so(4) whereby $\tau \in$ so(4) is mapped into its dual, $^*\tau$. The even subspace under * of so(4) corresponds to self-dual tensors, and, say, the first su(2) in the direct sum; the odd subspace corresponds to anti-self-dual tensors, and the second su(2) in the direct sum. A basis for so(4) may be chosen as follows. Each of the six skew-symmetric γ matrices has the property that the square of the matrix is equal to $-\gamma_0$. By Eq. (1), these six matrices are given by $(h = 1,2,3)$,

In[26]:=

$$2s^h = (\gamma^{23}, \gamma^{31}, \gamma^{21}), \tag{4}$$

and

$$2t^h = (\gamma^{45}, \gamma^{64}, \gamma^{65}). \tag{5}$$

Out[26]=

Under the involutive automorphism $\gamma^{AB} \to -\tilde{\gamma}^{AB}$ of so(3,3), the Lie algebra decomposes into the eigenvalue (-1) and eigenvalue $(+1)$ subspaces corresponding to, respectively, the nine linearly independent real traceless symmetric 4×4 matrices, and the six linearly independent real skew-symmetric 4×4 matrices. The eigenvalue $(+1)$ subspace is the subalgebra so(4), which is the Lie algebra of $\overline{\mathrm{SO}(4)}$, the maximal compact subgroup of $\overline{\mathrm{SL}(4,\mathbf{R})}$. The subalgebra $\mathrm{so}(4) \cong \mathrm{su}(2) + \mathrm{su}(2)$ may be further decomposed into the even (eigenvalue $+1$) and odd (eigenvalue -1) subspaces of the linear transformation of so(4) whereby $\tau \in \mathrm{so}(4)$ is mapped into its dual, $^*\tau$. The even subspace under * of so(4) corresponds to self-dual tensors, and, say, the first su(2) in the direct sum; the odd subspace corresponds to anti-self-dual tensors, and the second su(2) in the direct sum. A basis for so(4) may be chosen as follows. Each of the six skew-symmetric γ matrices has the property that the square of the matrix is equal to $-\gamma_0$. By Eq. (1), these six matrices are given by $(h = 1,2,3)$,

$$2s^h = (\gamma^{23}, \gamma^{31}, \gamma^{21}), \tag{4}$$

and

$$2t^h = (\gamma^{45}, \gamma^{64}, \gamma^{65}). \tag{5}$$

In[27]:=
```
Do[s4by4[h] = Table[Table[SelfDualAntiSymmetric[h, p, q], {q, 4}], {p, 4}],
 {h, 1, 3} ]
```

In[28]:=
```
Do[
 t4by4[h] = Table[Table[AntiSelfDualAntiSymmetric[h, p, q], {q, 4}], {p, 4}],
 {h, 1, 3} ]
```

In[29]:=
```
(*Protect[{s4by4[1],s4by4[2],s4by4[3],t4by4[1],t4by4[2],t4by4[3]}]*)
```

In[30]:=
```
Protect[s4by4, t4by4]
```

Out[30]=
```
{s4by4, t4by4}
```

Self Dual Anti-Symmetric

In[31]:=
```
Table[(1/2 Sum[(Sum[ϵ4[[p, q, j1, j2]] × s4by4[h][[j1, j2]], {j2, 1, 4}]), {j1, 1, 4}]) - s4by4[h][[p, q]],
 {h, 1, 3}, {q, 4}, {p, 4}]
```

Out[31]=
```
{{{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}},
 {{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}},
 {{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}}}
```

Anti SelfDual Anti-Symmetric

In[32]:=
```
Table[(1/2 Sum[(Sum[ϵ4[[p, q, j1, j2]] × t4by4[h][[j1, j2]], {j2, 1, 4}]), {j1, 1, 4}]) + t4by4[h][[p, q]],
 {h, 1, 3}, {q, 4}, {p, 4}]
```

Out[32]=
```
{{{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}},
 {{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}},
 {{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}}}
```

In[33]:=
```
{s4by4[#] // MatrixForm, MatrixForm[t4by4[#]]} & /@ Range[3]
```

Out[33]=

$$\left\{\left\{\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}\right\},\right.$$

$$\left.\left\{\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}\right\}, \left\{\begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}\right\}\right\}$$

The nine symmetric trace-free γ matrices may be denoted as $\gamma^{h'k}$, $h' = 1', 2', 3'$, where $1' = 6$, $2' = 5$, $3' = 4$. The $\gamma^{h'k}$ comprise a linearly independent basis for the nine-dimensional symmetric subspace of so(3,3). They may be ex-

In[34]:=
```
Table[{{J, K}, MatrixForm[s4by4[J].t4by4[K]]}, {J, 1, 3}, {K, 1, 3}]
```

Out[34]=

$$\left\{\left\{\left\{\{1,1\},\begin{pmatrix}1&0&0&0\\0&-1&0&0\\0&0&-1&0\\0&0&0&1\end{pmatrix}\right\},\left\{\{1,2\},\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}\right\},\left\{\{1,3\},\begin{pmatrix}0&0&1&0\\0&0&0&-1\\1&0&0&0\\0&-1&0&0\end{pmatrix}\right\}\right\},\right.$$

$$\left\{\left\{\{2,1\},\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&-1\\0&0&-1&0\end{pmatrix}\right\},\left\{\{2,2\},\begin{pmatrix}-1&0&0&0\\0&1&0&0\\0&0&-1&0\\0&0&0&1\end{pmatrix}\right\},\left\{\{2,3\},\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{pmatrix}\right\}\right\},$$

$$\left.\left\{\left\{\{3,1\},\begin{pmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{pmatrix}\right\},\left\{\{3,2\},\begin{pmatrix}0&0&0&-1\\0&0&1&0\\0&1&0&0\\-1&0&0&0\end{pmatrix}\right\},\left\{\{3,3\},\begin{pmatrix}-1&0&0&0\\0&-1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}\right\}\right\}\right\}$$

In[35]:=
```
Do[st[J, K] = s4by4[J].t4by4[K], {J, 1, 3}, {K, 1, 3}]
```

In[36]:=
```
Protect[st]
```

Out[36]=

{st}

In[37]:=
```
Table[{{J, K}, MatrixForm[st[J, K]]}, {J, 1, 3}, {K, 1, 3}]
```

Out[37]=

$$\left\{\left\{\left\{\{1,1\},\begin{pmatrix}1&0&0&0\\0&-1&0&0\\0&0&-1&0\\0&0&0&1\end{pmatrix}\right\},\left\{\{1,2\},\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}\right\},\left\{\{1,3\},\begin{pmatrix}0&0&1&0\\0&0&0&-1\\1&0&0&0\\0&-1&0&0\end{pmatrix}\right\}\right\},\right.$$

$$\left\{\left\{\{2,1\},\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&-1\\0&0&-1&0\end{pmatrix}\right\},\left\{\{2,2\},\begin{pmatrix}-1&0&0&0\\0&1&0&0\\0&0&-1&0\\0&0&0&1\end{pmatrix}\right\},\left\{\{2,3\},\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{pmatrix}\right\}\right\},$$

$$\left.\left\{\left\{\{3,1\},\begin{pmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{pmatrix}\right\},\left\{\{3,2\},\begin{pmatrix}0&0&0&-1\\0&0&1&0\\0&1&0&0\\-1&0&0&0\end{pmatrix}\right\},\left\{\{3,3\},\begin{pmatrix}-1&0&0&0\\0&-1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}\right\}\right\}\right\}$$

In[38]:=
```
Table[{{J, K}, MatrixForm[st[J, K] - Transpose[st[J, K]]]}, {J, 1, 3}, {K, 1, 3}]
```

Out[38]=

$$\left\{\left\{\left\{\{1,1\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\},\left\{\{1,2\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\},\left\{\{1,3\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\}\right\},\right.$$

$$\left\{\left\{\{2,1\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\},\left\{\{2,2\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\},\left\{\{2,3\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\}\right\},$$

$$\left.\left\{\left\{\{3,1\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\},\left\{\{3,2\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\},\left\{\{3,3\},\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}\right\}\right\}\right\}$$

In[39]:=
```
Flatten[{s4by4[#] & /@ Range[3], t4by4[#] & /@ Range[3],
  Flatten[Table[st[J, K], {J, 1, 3}, {K, 1, 3}], 1], {ID4}}, 1]
Length[%]
```

Out[39]=
```
{{{0, 0, 0, 1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {-1, 0, 0, 0}},
 {{0, 0, -1, 0}, {0, 0, 0, 1}, {1, 0, 0, 0}, {0, -1, 0, 0}},
 {{0, 1, 0, 0}, {-1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, -1, 0}},
 {{0, 0, 0, -1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {1, 0, 0, 0}},
 {{0, 0, -1, 0}, {0, 0, 0, -1}, {1, 0, 0, 0}, {0, 1, 0, 0}},
 {{0, 1, 0, 0}, {-1, 0, 0, 0}, {0, 0, 0, -1}, {0, 0, 1, 0}},
 {{1, 0, 0, 0}, {0, -1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, 1}},
 {{0, 1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, 1, 0}},
 {{0, 0, 1, 0}, {0, 0, 0, -1}, {1, 0, 0, 0}, {0, -1, 0, 0}},
 {{0, 1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, -1}, {0, 0, -1, 0}},
 {{-1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, 1}},
 {{0, 0, 0, 1}, {0, 0, 1, 0}, {0, 1, 0, 0}, {1, 0, 0, 0}},
 {{0, 0, 1, 0}, {0, 0, 0, 1}, {1, 0, 0, 0}, {0, 1, 0, 0}},
 {{0, 0, 0, -1}, {0, 0, 1, 0}, {0, 1, 0, 0}, {-1, 0, 0, 0}},
 {{-1, 0, 0, 0}, {0, -1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}},
 {{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}}}
```

Out[40]=
```
16
```

JOURNAL OF MATHEMATICAL PHYSICS VOLUME 4, NUMBER 7 JULY 1963

A Remarkable Representation of the 3 + 2 de Sitter Group

P. A. M. Dirac
Cambridge University, Cambridge, England and
*Institute for Advanced Study, Princeton, New Jersey**
(Received 20 February 1963)

In[41]:=
```
Unprotect[γABs];
γABs = Flatten[{s4by4[#] & /@ Range[3], t4by4[#] & /@ Range[3],
   Flatten[Table[st[J, K], {J, 1, 3}, {K, 1, 3}], 1], {ID4}}, 1]
```

Out[42]=
```
{{{0, 0, 0, 1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {-1, 0, 0, 0}},
 {{0, 0, -1, 0}, {0, 0, 0, 1}, {1, 0, 0, 0}, {0, -1, 0, 0}},
 {{0, 1, 0, 0}, {-1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, -1, 0}},
 {{0, 0, 0, -1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {1, 0, 0, 0}},
 {{0, 0, -1, 0}, {0, 0, 0, -1}, {1, 0, 0, 0}, {0, 1, 0, 0}},
 {{0, 1, 0, 0}, {-1, 0, 0, 0}, {0, 0, 0, -1}, {0, 0, 1, 0}},
 {{1, 0, 0, 0}, {0, -1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, 1}},
 {{0, 1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, 1, 0}},
 {{0, 0, 1, 0}, {0, 0, 0, -1}, {1, 0, 0, 0}, {0, -1, 0, 0}},
 {{0, 1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, -1}, {0, 0, -1, 0}},
 {{-1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, 1}},
 {{0, 0, 0, 1}, {0, 0, 1, 0}, {0, 1, 0, 0}, {1, 0, 0, 0}},
 {{0, 0, 1, 0}, {0, 0, 0, 1}, {1, 0, 0, 0}, {0, 1, 0, 0}},
 {{0, 0, 0, -1}, {0, 0, 1, 0}, {0, 1, 0, 0}, {-1, 0, 0, 0}},
 {{-1, 0, 0, 0}, {0, -1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}},
 {{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}}}
```

In[43]:=
```
Length[γABs]
```

Out[43]=
```
16
```

In[44]:=
```
Protect[γABs]
```

Out[44]=
```
{γABs}
```

In[45]:=
```
γABs[[1]]
```

Out[45]=
```
{{0, 0, 0, 1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {-1, 0, 0, 0}}
```

In[46]:=
```
γABs[[-1]]
```

Out[46]=
```
{{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}}
```

γ matrices

In[47]:=
```
g3 = DiagonalMatrix[{1, 1, -1}]
```
Out[47]=
```
{{1, 0, 0}, {0, 1, 0}, {0, 0, -1}}
```

In[48]:=
```
Do[Do[Gr[h, k] = t4by4[h].s4by4[k] * (-g3[[h, h]] * g3[[k, k]]), {k, 1, 3}], {h, 1, 3}]
```

In[49]:=
```
Gr[1, 1] // MatrixForm
```
Out[49]//MatrixForm=

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

In[50]:=
```
Unprotect[γ];
```
WAIT until γ^5 to Protect[γ]

In[51]:=
```
γ = {Gr[1, 1], Gr[1, 2], Gr[1, 3], t4by4[2]};
```

In[52]:=
```
Table[FullSimplify[γ[[h]].γ[[h]] === g_αβ[[h, h]] * ID4], {h, 1, 4}]
```
Out[52]=
```
{True, True, True, True}
```

$$\gamma^{\alpha}\gamma^{\beta} + \gamma^{\beta}\gamma^{\alpha} = 2\gamma_0 g^{\alpha\beta}, \tag{4}$$

where

$$g^{\alpha\beta} = g_{\alpha\beta} = \mathrm{diag}(1,1,1,-1) \tag{5}$$

and γ_0 denotes the 4×4 unit matrix. The $\Sigma^{\alpha\beta}$ of Eq. (1)

In[53]:=
```
Table[{{h, k}, FullSimplify[(γ[[h]].γ[[k]] + γ[[k]].γ[[h]])/2] === g_αβ[[h, k]] ID4},
  {h, 1, 4}, {k, 1, 4}] // MatrixForm
```
Out[53]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{2,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{3,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{4,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,4\} \\ \text{True} \end{pmatrix} \end{pmatrix}$$

In[54]:=
```
Table[{{h, k}, FullSimplify[(γ[[h]].γ[[k]] + γ[[k]].γ[[h]])/2] === η4488[[1 + h, 1 + k]] ID4},
  {h, 1, 4}, {k, 1, 4}] // MatrixForm
```

Out[54]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{2,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{3,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{4,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,4\} \\ \text{True} \end{pmatrix} \end{pmatrix}$$

$$\gamma^5 = -\gamma^1 \gamma^2 \gamma^3 \gamma^4$$

In[55]:=
```
-γ[[1]].γ[[2]].γ[[3]].γ[[4]]
```

Out[55]=
```
{{0, -1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, -1, 0}}
```

In[56]:=
```
AppendTo[γ, (-γ[[1]].γ[[2]].γ[[3]].γ[[4]])]
```

Out[56]=
```
{{{-1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, -1}},
 {{0, -1, 0, 0}, {-1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, 1, 0}},
 {{0, 0, 1, 0}, {0, 0, 0, 1}, {1, 0, 0, 0}, {0, 1, 0, 0}},
 {{0, 0, -1, 0}, {0, 0, 0, -1}, {1, 0, 0, 0}, {0, 1, 0, 0}},
 {{0, -1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, -1, 0}}}
```

In[57]:=
```
Protect[γ]
```

Out[57]=
```
{γ}
```

In[58]:=
```
γ[[5]]
```

Out[58]=
```
{{0, -1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, -1, 0}}
```

In[59]:=
```
γ[[5]] === -t4by4[3]
```

Out[59]=
```
True
```

In[60]:=
```
Table[
  {{h, k}, FullSimplify[γ⟦h⟧.γ⟦k⟧ + γ⟦k⟧.γ⟦h⟧] === 2 * η4488⟦1 + h, 1 + k⟧ ID4},
  {h, 1, 5}, {k, 1, 5}] // MatrixForm
```

Out[60]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{2,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{3,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{4,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{5,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,5\} \\ \text{True} \end{pmatrix} \end{pmatrix}$$

$$\gamma^{\alpha} = \gamma^{\alpha 6}, \tag{19}$$

$$\gamma^{5} = \gamma^{56}, \tag{20}$$

$$\gamma^{\alpha}\gamma^{5} = \gamma^{\alpha 5}, \tag{21}$$

$$\epsilon = \gamma^{64}.$$

In[61]:=
```
-γ⟦4⟧ // MatrixForm
```

Out[61]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

In[62]:=
```
sϵ = {ϵ → t4by4[2]}
```

Out[62]=
```
{ϵ → {{0, 0, -1, 0}, {0, 0, 0, -1}, {1, 0, 0, 0}, {0, 1, 0, 0}}}
```

In[63]:=
```
Transpose[ϵ /. sϵ] === (-ϵ /. sϵ)
```

Out[63]=
```
True
```

In[64]:=
```
Table[{{A1}, FullSimplify[(ϵ /. sϵ).γ⟦A1⟧ === Transpose[(ϵ /. sϵ).γ⟦A1⟧]]},
 {A1, 1, 5}]
```

Out[64]=
```
{{{1}, True}, {{2}, True}, {{3}, True}, {{4}, True}, {{5}, False}}
```

In[65]:=
```
γ⟦4⟧ === t4by4[2]
```

Out[65]=
```
True
```

In[66]:=
```
t4by4[1]
```
Out[66]=
```
{{0, 0, 0, -1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {1, 0, 0, 0}}
```

γ5, ...

In[67]:=
```
γ[[5]]
```
Out[67]=
```
{{0, -1, 0, 0}, {1, 0, 0, 0}, {0, 0, 0, 1}, {0, 0, -1, 0}}
```

In[68]:=
```
γ[[5]] === -t4by4[3]
```
Out[68]=
```
True
```

In[69]:=
```
γ[[4]] === t4by4[2]
```
Out[69]=
```
True
```

In[70]:=
```
γ[[4]] // MatrixForm
```
Out[70]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

In[71]:=
```
sϵ = {ϵ → t4by4[2]}
```
Out[71]=
```
{ϵ → {{0, 0, -1, 0}, {0, 0, 0, -1}, {1, 0, 0, 0}, {0, 1, 0, 0}}}
```

In[72]:=
```
Transpose[ϵ /. sϵ] === (-ϵ /. sϵ)
```
Out[72]=
```
True
```

In[73]:=
```
(*Parallel*)Table[{{A1}, FullSimplify[
   (ϵ /. sϵ).γ[[A1]] === Transpose[(ϵ /. sϵ).γ[[A1]]]]}, {A1, 1, 5}]
```
Out[73]=
```
{{{1}, True}, {{2}, True}, {{3}, True}, {{4}, True}, {{5}, False}}
```

In[74]:=
```
γ[[4]] === t4by4[2]
```
Out[74]=
```
True
```

In[75]:=
```
t4by4[1]
```

Out[75]=
```
{{0, 0, 0, -1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {1, 0, 0, 0}}
```

In[76]:=
```
(*γ⟦5⟧===γs⟦5,6⟧*)
```

In[77]:=
```
Table[{{h, k}, FullSimplify[γ⟦h⟧.γ⟦k⟧ + γ⟦k⟧.γ⟦h⟧] ===
    2 * η4488⟦1 + h, 1 + k⟧ ID4}, {h, 1, 5}, {k, 1, 5}] // MatrixForm
```

Out[77]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{2,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{3,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{4,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,5\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{5,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,4\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{5,5\} \\ \text{True} \end{pmatrix} \end{pmatrix}$$

In[78]:=
```
(γ56 = -γ⟦5⟧) // MatrixForm
(γ64 = -γ⟦4⟧) // MatrixForm
(*(γ45=t4by4[2])//MatrixForm*)
(γ45 = γ⟦1⟧.γ⟦2⟧.γ⟦3⟧) // MatrixForm
```

Out[78]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Out[79]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

Out[80]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

In[81]:=
```
γ456 = {γ45, γ56, γ64}
```

Out[81]=
```
{{{0, 0, 0, 1}, {0, 0, -1, 0}, {0, 1, 0, 0}, {-1, 0, 0, 0}},
 {{0, 1, 0, 0}, {-1, 0, 0, 0}, {0, 0, 0, -1}, {0, 0, 1, 0}},
 {{0, 0, 1, 0}, {0, 0, 0, 1}, {-1, 0, 0, 0}, {0, -1, 0, 0}}}
```

In[82]:=
```
Table[{{j, k}, (γ456[[j]].γ456[[k]] + γ456[[k]].γ456[[j]])/2 ===
    If[j == k, -ID4, Zero4]}, {j, 1, 3}, {k, 1, 3}] // MatrixForm
```

Out[82]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1, 1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1, 2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1, 3\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{2, 1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2, 2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2, 3\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{3, 1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3, 2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3, 3\} \\ \text{True} \end{pmatrix} \end{pmatrix}$$

Dirac Identity; generalized by Nash

JOURNAL OF MATHEMATICAL PHYSICS VOLUME 4, NUMBER 7 JULY 1963

A Remarkable Representation of the 3 + 2 de Sitter Group

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Cambridge University, Cambridge, England and
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(Received 20 February 1963)

SOME PROPERTIES OF THE OPERATORS χ

Lemma. For any symmetrical 4×4 matrix S,

$$\gamma_{12}S\gamma_{12} + \gamma_{23}S\gamma_{23} + \gamma_{31}S\gamma_{31} = S - \langle S\rangle\gamma_0, \qquad (22)$$

and similarly,

$$\gamma_{45}S\gamma_{45} + \gamma_{56}S\gamma_{56} + \gamma_{64}S\gamma_{64} = S - \langle S\rangle\gamma_0. \qquad (23)$$

In[ ]:=

To prove (22), we note that any symmetric γ_{AB} has one of its suffixes equal to 1, 2, or 3 and the other equal to 4, 5, or 6 and therefore it commutes with one of the three quantities γ_{12}, γ_{23}, γ_{31} and anticommutes with the other two. Thus if such a γ_{AB} is substituted for S in (22), two of the terms on the left become S and the third one $-S$, and the $\langle S\rangle$ on the right is zero. Also (22) holds with $S = \gamma_0$, so it holds for any symmetric S. The proof of (23) is similar.

Identities satisfied by the generators of the Dirac algebra

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(Received 26 July 1983; accepted for publication 23 September 1983)

3. A BASIC LEMMA

Lemma: Let X be an arbitrary 4×4 matrix; then

$$\gamma^{56}X\gamma^{56}+\gamma^{64}\tilde{X}\gamma^{64}+\gamma^{45}\tilde{X}\gamma^{45} = X-\gamma_0\,\mathrm{tr}\,X+\gamma^{56}\,\mathrm{tr}\,\gamma^{56}X, \tag{45}$$

where $\tilde{X}$ denotes the transpose of X, and tr X is the trace of X. This identity is valid for any cyclic permutation of $(\gamma^{56},\gamma^{64},\gamma^{45})$, and under the replacement $\gamma^{56}\to\gamma^{12}$, $\gamma^{64}\to\gamma^{31}$, and $\gamma^{45}\to\gamma^{23}$.

Proof: Eq. (45) is linear in X; we verify that this equation is true for $X=\gamma_0$, γ^{56}, $\gamma^{\alpha 5}$, $\gamma^{\alpha 6}$, and $\gamma^{\alpha\beta}$. Note that only for $X=\gamma_0$ (resp γ^{56}) is tr X (resp tr $\gamma^{56}X$) nonvanishing.

(i) $X=\gamma_0$; since $(\gamma^{56})^2=-\gamma_0=(\gamma^{64})^2=(\gamma^{45})^2$, Eq. (45) yields

$$(\gamma^{56})^2+(\gamma^{64})^2+(\gamma^{45})^2=-3\gamma_0 = \gamma_0-\gamma_0\,\mathrm{tr}\,\gamma_0+\gamma^{56}\,\mathrm{tr}\,\gamma^{56}=\gamma_0-4\gamma_0;$$

BELOW:

anti4by4[[#]][[2]] & /@ {1, 5, 6}===={{{5}, {4}↔γ64}, {{12}, {1, 2, 3}}, {{16}, {1, 2, 3, 4}↔γ56}}

BE VERY CAREFULL :

```
{ γ64 ,        γ45 ,            γ56 } :
{ γ[[4]] , γ[[1]].γ[[2]].γ[[3]] γ[[1]].γ[[2]].γ[[3]].γ[[4]] }

{ {4} ,    {1, 2, 3},        x5, x6, x7} :
```

{{{5}, {4}}, {{12}, {1, 2, 3}}, {{16}, {1, 2, 3, 4}}}

In[ ]:= anti4by4[[#]][[2]] & /@ {2, 3, 4}

Out[ ]=

{{{6}, {1, 2}}, {{7}, {1, 3}}, {{9}, {2, 3}}}

In[83]:=
```
Clear[DiracNash];
DiracNash[M_] :=
 Zero4 === FullSimplify[γ56.M.γ56 + γ64.Transpose[M].γ64 +
    γ45.Transpose[M].γ45 - M + ID4 * Tr[M] - γ56 * Tr[γ56.M]]
```

In[85]:=
```
DiracNash[#] & /@ {γ56, γ64, γ45}
```
Out[85]=
```
{True, True, True}
```

In[86]:=
```
Table[DiracNash[γ[[j]]], {j, 1, 5}]
```
Out[86]=
```
{True, True, True, True, True}
```

In[87]:=
```
DiracNash[#] & /@ γABs // Union
```
Out[87]=
```
{True}
```

$$S^{\alpha\beta} = -\tfrac{1}{4}[\gamma^{\alpha}, \gamma^{\beta}]. \qquad (3)$$

In[88]:=
```
S44αβ = -1/4 Table[FullSimplify[γ[[h]].γ[[k]] - γ[[k]].γ[[h]]], {h, 1, 4}, {k, 1, 4}];
```

In[89]:=
```
Protect[S44αβ]
```
Out[89]=
```
{S44αβ}
```

In[90]:=
```
η44 = DiagonalMatrix[{1, 1, 1, -1}]
```
Out[90]=
```
{{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, -1}}
```

In[91]:=
```
Protect[η44]
```
Out[91]=
```
{η44}
```

S44αβ commutation relations; misc :

$$S^{\alpha\beta} = -\tfrac{1}{4}[\gamma^{\alpha},\gamma^{\beta}], \tag{22}$$

$$S^{\alpha\beta} = -\tfrac{1}{2}\gamma^{\alpha\beta}. \tag{23}$$

The symplectic form ϵ on D_4 is defined to be

$$\epsilon = \gamma^{64}. \tag{24}$$

As a consequence of Eq. (1), and the definitions of Eqs. (16)–(24), are the identities

$$\tilde{\gamma}^{\alpha}\epsilon = -\epsilon\gamma^{\alpha}, \tag{25}$$

$$\tilde{S}^{\alpha\beta}\epsilon = -\epsilon S^{\alpha\beta}, \tag{26}$$

$$[S^{\alpha\beta},\gamma_{\mu}] = \delta^{\alpha}_{\mu}\gamma^{\beta} - \delta^{\beta}_{\mu}\gamma^{\alpha}, \tag{27}$$

$$[S^{\alpha\beta},S^{\mu\nu}] = g^{\alpha\mu}S^{\beta\nu} - g^{\alpha\nu}S^{\beta\mu} - g^{\beta\mu}S^{\alpha\nu} + g^{\beta\nu}S^{\alpha\mu}, \tag{28}$$

and

$$\gamma^{5}S^{\alpha\beta} = \tfrac{1}{2}g^{\alpha\mu}g^{\beta\nu}\epsilon_{\mu\nu\lambda\sigma}S^{\lambda\sigma}. \tag{29}$$

```
S44αβ = -1/4 Table[FullSimplify[γ[[h]].γ[[k]] - γ[[k]].γ[[h]]], {h, 1, 4}, {k, 1, 4}];
```

In[92]:=
```
(*SAB=
  Table[1/4 (T16^A[A1].T16^A[B1]-T16^A[B1].T16^A[A1]), {A1,0, 7},{B1,0, 7}];*)
```

In[93]:=
```
(*ParallelTable[
   FullSimplify[σ16.SAB[[A1,B1]]===-Transpose[σ16.SAB[[A1,B1]]]],
   {A1,1, 8},{B1,1,8}]//Flatten//Union*)
```

In[94]:=
```
(*ParallelTable[
 {{A1,B1},FullSimplify[SAB[[A1,B1]]===-Transpose[SAB[[A1,B1]]]]},
 {A1,1, 8},{B1,1,8}]*)
```

In[95]:=
```
ParallelTable[
   FullSimplify[S44αβ[[A1, B1]].S44αβ[[A2, B2]] - S44αβ[[A2, B2]].S44αβ[[A1, B1]] ==
     (η44[[A1, A2]] × S44αβ[[B1, B2]] - η44[[A1, B2]] × S44αβ[[B1, A2]] -
       η44[[B1, A2]] × S44αβ[[A1, B2]] + η44[[B1, B2]] × S44αβ[[A1, A2]])], {A1, 1, 3},
   {B1, A1 + 1, 4}, {A2, 1, 3}, {B2, A2 + 1, 4}] // Flatten // Union
```

Out[95]=
```
{True}
```

In[96]:=
```
ParallelTable[{{A1, B1, A2, B2},
   FullSimplify[S44αβ⟦A1, B1⟧.S44αβ⟦A2, B2⟧ - S44αβ⟦A2, B2⟧.S44αβ⟦A1, B1⟧ ==
     (η44⟦A1, A2⟧ × S44αβ⟦B1, B2⟧ - η44⟦A1, B2⟧ × S44αβ⟦B1, A2⟧ -
       η44⟦B1, A2⟧ × S44αβ⟦A1, B2⟧ + η44⟦B1, B2⟧ × S44αβ⟦A1, A2⟧)]},
  {A1, 1, 4}, {B1, 1, 4}, {A2, 1, 4}, {B2, 1, 4}] // MatrixForm
```

Out[96]//MatrixForm=

{1, 1, 1, 1} True	{1, 1, 1, 2} True	{1, 1, 1, 3} True	{1, 1, 1, 4} True	{1, 2, 1, 1} True	{1,
{1, 1, 2, 1} True	{1, 1, 2, 2} True	{1, 1, 2, 3} True	{1, 1, 2, 4} True	{1, 2, 2, 1} True	{1,
{1, 1, 3, 1} True	{1, 1, 3, 2} True	{1, 1, 3, 3} True	{1, 1, 3, 4} True	{1, 2, 3, 1} True	{1,
{1, 1, 4, 1} True	{1, 1, 4, 2} True	{1, 1, 4, 3} True	{1, 1, 4, 4} True	{1, 2, 4, 1} True	{1,
{2, 1, 1, 1} True	{2, 1, 1, 2} True	{2, 1, 1, 3} True	{2, 1, 1, 4} True	{2, 2, 1, 1} True	{2,
{2, 1, 2, 1} True	{2, 1, 2, 2} True	{2, 1, 2, 3} True	{2, 1, 2, 4} True	{2, 2, 2, 1} True	{2,
{2, 1, 3, 1} True	{2, 1, 3, 2} True	{2, 1, 3, 3} True	{2, 1, 3, 4} True	{2, 2, 3, 1} True	{2,
{2, 1, 4, 1} True	{2, 1, 4, 2} True	{2, 1, 4, 3} True	{2, 1, 4, 4} True	{2, 2, 4, 1} True	{2,
{3, 1, 1, 1} True	{3, 1, 1, 2} True	{3, 1, 1, 3} True	{3, 1, 1, 4} True	{3, 2, 1, 1} True	{3,
{3, 1, 2, 1} True	{3, 1, 2, 2} True	{3, 1, 2, 3} True	{3, 1, 2, 4} True	{3, 2, 2, 1} True	{3,
{3, 1, 3, 1} True	{3, 1, 3, 2} True	{3, 1, 3, 3} True	{3, 1, 3, 4} True	{3, 2, 3, 1} True	{3,
{3, 1, 4, 1} True	{3, 1, 4, 2} True	{3, 1, 4, 3} True	{3, 1, 4, 4} True	{3, 2, 4, 1} True	{3,
{4, 1, 1, 1} True	{4, 1, 1, 2} True	{4, 1, 1, 3} True	{4, 1, 1, 4} True	{4, 2, 1, 1} True	{4,
{4, 1, 2, 1} True	{4, 1, 2, 2} True	{4, 1, 2, 3} True	{4, 1, 2, 4} True	{4, 2, 2, 1} True	{4,
{4, 1, 3, 1} True	{4, 1, 3, 2} True	{4, 1, 3, 3} True	{4, 1, 3, 4} True	{4, 2, 3, 1} True	{4,
{4, 1, 4, 1} True	{4, 1, 4, 2} True	{4, 1, 4, 3} True	{4, 1, 4, 4} True	{4, 2, 4, 1} True	{4,

In[97]:=
```
ParallelTable[FullSimplify[S44αβ⟦A1, B1⟧.γ⟦B2⟧ - γ⟦B2⟧.S44αβ⟦A1, B1⟧ ==
     (η44⟦B2, A1⟧ × γ⟦B1⟧ - η44⟦B2, B1⟧ × γ⟦A1⟧)],
   {A1, 1, 4}, {B1, 1, 4}, {B2, 1, 4}] // Flatten // Union
```

Out[97]=

{True}

In[98]:=
```
ParallelTable[
  {{A1, B1, B2}, FullSimplify[S44αβ⟦A1, B1⟧.γ⟦B2⟧ - γ⟦B2⟧.S44αβ⟦A1, B1⟧ ==
     (η44⟦B2, A1⟧ × γ⟦B1⟧ - η44⟦B2, B1⟧ × γ⟦A1⟧)]},
  {A1, 1, 4}, {B1, 1, 4}, {B2, 1, 4}] // MatrixForm
```

Out[98]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1,1,1\} & \text{True} \\ \{1,1,2\} & \text{True} \\ \{1,1,3\} & \text{True} \\ \{1,1,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{1,2,1\} & \text{True} \\ \{1,2,2\} & \text{True} \\ \{1,2,3\} & \text{True} \\ \{1,2,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{1,3,1\} & \text{True} \\ \{1,3,2\} & \text{True} \\ \{1,3,3\} & \text{True} \\ \{1,3,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{1,4,1\} & \text{True} \\ \{1,4,2\} & \text{True} \\ \{1,4,3\} & \text{True} \\ \{1,4,4\} & \text{True} \end{pmatrix} \\ \begin{pmatrix} \{2,1,1\} & \text{True} \\ \{2,1,2\} & \text{True} \\ \{2,1,3\} & \text{True} \\ \{2,1,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{2,2,1\} & \text{True} \\ \{2,2,2\} & \text{True} \\ \{2,2,3\} & \text{True} \\ \{2,2,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{2,3,1\} & \text{True} \\ \{2,3,2\} & \text{True} \\ \{2,3,3\} & \text{True} \\ \{2,3,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{2,4,1\} & \text{True} \\ \{2,4,2\} & \text{True} \\ \{2,4,3\} & \text{True} \\ \{2,4,4\} & \text{True} \end{pmatrix} \\ \begin{pmatrix} \{3,1,1\} & \text{True} \\ \{3,1,2\} & \text{True} \\ \{3,1,3\} & \text{True} \\ \{3,1,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{3,2,1\} & \text{True} \\ \{3,2,2\} & \text{True} \\ \{3,2,3\} & \text{True} \\ \{3,2,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{3,3,1\} & \text{True} \\ \{3,3,2\} & \text{True} \\ \{3,3,3\} & \text{True} \\ \{3,3,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{3,4,1\} & \text{True} \\ \{3,4,2\} & \text{True} \\ \{3,4,3\} & \text{True} \\ \{3,4,4\} & \text{True} \end{pmatrix} \\ \begin{pmatrix} \{4,1,1\} & \text{True} \\ \{4,1,2\} & \text{True} \\ \{4,1,3\} & \text{True} \\ \{4,1,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{4,2,1\} & \text{True} \\ \{4,2,2\} & \text{True} \\ \{4,2,3\} & \text{True} \\ \{4,2,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{4,3,1\} & \text{True} \\ \{4,3,2\} & \text{True} \\ \{4,3,3\} & \text{True} \\ \{4,3,4\} & \text{True} \end{pmatrix} & \begin{pmatrix} \{4,4,1\} & \text{True} \\ \{4,4,2\} & \text{True} \\ \{4,4,3\} & \text{True} \\ \{4,4,4\} & \text{True} \end{pmatrix} \end{pmatrix}$$

In[99]:=
```
ParallelTable[{{A1, B1}, FullSimplify[(ϵ /. sϵ).S44αβ⟦A1, B1⟧ ===
    Transpose[(ϵ /. sϵ).S44αβ⟦A1, B1⟧]]}, {A1, 1, 3}, {B1, A1 + 1, 4}]
```

Out[99]=
```
{{{{1, 2}, True}, {{1, 3}, True}, {{1, 4}, True}},
 {{{2, 3}, True}, {{2, 4}, True}}, {{{3, 4}, True}}}
```

another real BASIS of 4 × 4 matrices , using PauliMatrix[]:

In[100]:=
```
σ22 =
 Flatten[{{IdentityMatrix[2]}, If[# == 2, ⅈ, 1] * PauliMatrix[#] & /@ Range[3]}, 1]
```

Out[100]=
```
{{{1, 0}, {0, 1}}, {{0, 1}, {1, 0}}, {{0, 1}, {-1, 0}}, {{1, 0}, {0, -1}}}
```

In[101]:=
```
σ̄22 = Flatten[
  {{-IdentityMatrix[2]}, If[# == 2, ⅈ, 1] * PauliMatrix[#] & /@ Range[3]}, 1]
```

Out[101]=
```
{{{-1, 0}, {0, -1}}, {{0, 1}, {1, 0}}, {{0, 1}, {-1, 0}}, {{1, 0}, {0, -1}}}
```

In[102]:=
```
-IdentityMatrix[2] === Dot @@ (σ22⟦#⟧ & /@ Range[Length[σ22]])
```

Out[102]=
```
True
```

In[103]:=

```
Table[{{A, B}, FullSimplify[1/2 Tr[1/2 (σ22[[A]].σ22̄[[B]] + σ22[[B]].σ22̄[[A]])]]},
  {A, 1, 4}, {B, 1, 4}] // MatrixForm
```

Out[103]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1,1\} \\ -1 \end{pmatrix} & \begin{pmatrix} \{1,2\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{1,3\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{1,4\} \\ 0 \end{pmatrix} \\ \begin{pmatrix} \{2,1\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{2,2\} \\ 1 \end{pmatrix} & \begin{pmatrix} \{2,3\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{2,4\} \\ 0 \end{pmatrix} \\ \begin{pmatrix} \{3,1\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{3,2\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{3,3\} \\ -1 \end{pmatrix} & \begin{pmatrix} \{3,4\} \\ 0 \end{pmatrix} \\ \begin{pmatrix} \{4,1\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{4,2\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{4,3\} \\ 0 \end{pmatrix} & \begin{pmatrix} \{4,4\} \\ 1 \end{pmatrix} \end{pmatrix}$$

In[104]:=

```
η2244 = DiagonalMatrix[{-1, 1, -1, 1}]
```

Out[104]=

```
{{-1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, 1}}
```

In[105]:=

```
Table[{{A, B}, FullSimplify[ExpandAll[
      1/2 (σ22[[A]].σ22̄[[B]] + σ22[[B]].σ22̄[[A]]) == η2244[[A, B]] * IdentityMatrix[2]]]},
  {A, 1, 4}, {B, 1, 4}] // MatrixForm
```

Out[105]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} \{1,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{1,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{2,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{2,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{3,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{3,4\} \\ \text{True} \end{pmatrix} \\ \begin{pmatrix} \{4,1\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,2\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,3\} \\ \text{True} \end{pmatrix} & \begin{pmatrix} \{4,4\} \\ \text{True} \end{pmatrix} \end{pmatrix}$$

In[106]:=

```
Do[yyy[j] = ArrayFlatten[{{0, σ22[[j]]}, {-σ22[[j]], 0}}], {j, 4}];
yyy[5] = ArrayFlatten[{{0, IdentityMatrix[2]}, {IdentityMatrix[2], 0}}]
```

Out[106]=

```
{{0, 0, 1, 0}, {0, 0, 0, 1}, {1, 0, 0, 0}, {0, 1, 0, 0}}
```

In[107]:=

```
yyy[5] = {{0, 0, -1, 0}, {0, 0, 0, 1}, {-1, 0, 0, 0}, {0, 1, 0, 0}}
```

Out[107]=

```
{{0, 0, -1, 0}, {0, 0, 0, 1}, {-1, 0, 0, 0}, {0, 1, 0, 0}}
```

In[108]:=

```
(yyy[6] = Dot @@ (yyy[#] & /@ Range[5])) // MatrixForm
```

Out[108]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

In[109]:=

```
MatrixForm[yyy[#]] & /@ Range[6]
```

Out[109]=

$$\left\{\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}\right\}$$

In[110]:=

```
Do[xxx[j] = ArrayFlatten[{{0, σ̄22[[j]]}, {σ22[[j]], 0}}], {j, 4}]
```

In[111]:=

```
MatrixForm[xxx[#]] & /@ Range[4]
```

Out[111]=

$$\left\{\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}\right\}$$

In[112]:=

```
xxx[5] = Dot @@ (xxx[#] & /@ Range[4])
```

Out[112]=

```
{{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, -1}}
```

In[113]:=

```
txA0 = {{IdentityMatrix[4], {0}}};
Length[txA0]
txA = {}; Do[AppendTo[txA, {xxx[j], {j}}], {j, 1, 4}]
Length[txA]
txAB = {};
Do[AppendTo[txAB, {xxx[j].xxx[k], {j, k}}], {j, 1, 3}, {k, j + 1, 4}]
Length[txAB]
txABC = {};
Do[AppendTo[txABC, {xxx[j].xxx[k].xxx[h], {j, k, h}}],
 {j, 1, 2}, {k, j + 1, 3}, {h, k + 1, 4}];
Length[txABC]
txABCD = {{xxx[1].xxx[2].xxx[3].xxx[4], {1, 2, 3, 4}}};
Length[txABCD]
```

Out[114]=

1

Out[116]=

4

Out[118]=

6

Out[120]=

4

Out[122]=

1

In[123]:=

```
base4by4 = Flatten[{txA0, txA, txAB, txABC, txABCD}, 1]; Length[base4by4]
```

Out[123]=

16

In[124]:=

```
DiracNash[#[[1]]] & /@ base4by4 // Union
```

Out[124]=

{True}

In[125]:=

```
Table[{{A, B}, Block[{r}, r = FullSimplify[1/4 Tr[
        1/2 (base4by4[[A, 1]].base4by4[[B, 1]] + base4by4[[B, 1]].base4by4[[A, 1]])]];
    If[r == 0, Style[r, Red], Style[r, Darker[Green]]]]},
  {A, 1, Length[base4by4]}, {B, 1, Length[base4by4]}] // MatrixForm
```

Out[125]//MatrixForm=

{1, 1} 1	{1, 2} 0	{1, 3} 0	{1, 4} 0	{1, 5} 0	{1, 6} 0	{1, 7} 0	{1, 8} 0
{2, 1} 0	{2, 2} -1	{2, 3} 0	{2, 4} 0	{2, 5} 0	{2, 6} 0	{2, 7} 0	{2, 8} 0
{3, 1} 0	{3, 2} 0	{3, 3} 1	{3, 4} 0	{3, 5} 0	{3, 6} 0	{3, 7} 0	{3, 8} 0
{4, 1} 0	{4, 2} 0	{4, 3} 0	{4, 4} -1	{4, 5} 0	{4, 6} 0	{4, 7} 0	{4, 8} 0
{5, 1} 0	{5, 2} 0	{5, 3} 0	{5, 4} 0	{5, 5} 1	{5, 6} 0	{5, 7} 0	{5, 8} 0
{6, 1} 0	{6, 2} 0	{6, 3} 0	{6, 4} 0	{6, 5} 0	{6, 6} 1	{6, 7} 0	{6, 8} 0
{7, 1} 0	{7, 2} 0	{7, 3} 0	{7, 4} 0	{7, 5} 0	{7, 6} 0	{7, 7} -1	{7, 8} 0
{8, 1} 0	{8, 2} 0	{8, 3} 0	{8, 4} 0	{8, 5} 0	{8, 6} 0	{8, 7} 0	{8, 8} 1
{9, 1} 0	{9, 2} 0	{9, 3} 0	{9, 4} 0	{9, 5} 0	{9, 6} 0	{9, 7} 0	{9, 8} 0
{10, 1} 0	{10, 2} 0	{10, 3} 0	{10, 4} 0	{10, 5} 0	{10, 6} 0	{10, 7} 0	{10, 8} 0
{11, 1} 0	{11, 2} 0	{11, 3} 0	{11, 4} 0	{11, 5} 0	{11, 6} 0	{11, 7} 0	{11, 8} 0
{12, 1} 0	{12, 2} 0	{12, 3} 0	{12, 4} 0	{12, 5} 0	{12, 6} 0	{12, 7} 0	{12, 8} 0
{13, 1} 0	{13, 2} 0	{13, 3} 0	{13, 4} 0	{13, 5} 0	{13, 6} 0	{13, 7} 0	{13, 8} 0
{14, 1} 0	{14, 2} 0	{14, 3} 0	{14, 4} 0	{14, 5} 0	{14, 6} 0	{14, 7} 0	{14, 8} 0
{15, 1} 0	{15, 2} 0	{15, 3} 0	{15, 4} 0	{15, 5} 0	{15, 6} 0	{15, 7} 0	{15, 8} 0
{16, 1} 0	{16, 2} 0	{16, 3} 0	{16, 4} 0	{16, 5} 0	{16, 6} 0	{16, 7} 0	{16, 8} 0

In[126]:=

```
base4by4[[16]][[1]]
```

Out[126]=

```
{{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, -1}}
```

In[127]:=

```
anti4by4 = {};
Do[If[base4by4[[k, 1]] == -Transpose[base4by4[[k, 1]]], AppendTo[anti4by4,
   {base4by4[[k, 1]], {{k}, base4by4[[k, 2]]}}] ], {k, 1, Length[base4by4]}];
Length[anti4by4]
anti4by4[[%]]
```

Out[128]=

6

Out[129]=

{{{0, 0, 0, 1}, {0, 0, 1, 0}, {0, -1, 0, 0}, {-1, 0, 0, 0}}, {{14}, {1, 3, 4}}}

In[130]:=

```
symm4by4 = {};
Do[If[base4by4[[k, 1]] == Transpose[base4by4[[k, 1]]], AppendTo[symm4by4,
   {base4by4[[k, 1]], {{k}, base4by4[[k, 2]]}}] ], {k, 1, Length[base4by4]}];
Length[symm4by4]
symm4by4[[%]]
```

Out[131]=

10

Out[132]=

{{{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, -1}}, {{16}, {1, 2, 3, 4}}}

In[133]:=

```
η16 =
 DiagonalMatrix[Table[FullSimplify[1/4 Tr[symm4by4[[A, 1]].symm4by4[[A, 1]]]],
   {A, Length[symm4by4]}]];
Tr[η16]
```

Out[133]=

10

A new spin-$\frac{1}{2}$ wave equation

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(Received 24 January 1984; accepted for publication 16 November 1984)

the description of a (composite) massive spin-$\frac{1}{2}$ particle. In this model, the spin degrees of freedom of the particle are carried by the functional dependence of the wave function on a real spinor independent variable u. Here, u is an element of an internal space D_4, real four-dimensional Dirac space. The wave equation itself is cast on $M_4 \times D_4$, where M_4 denotes Minkowski space-time. Solutions of the wave equation, when restricted to D_4, belong to the Hilbert space $L^2(D_4)$ of square-integrable smooth functions on D_4.

$$\tfrac{1}{2}\Sigma^{\alpha\beta}\Sigma_{\alpha\beta} = j_0^2 - c^2 - 1 \tag{6}$$

and

$$\tfrac{1}{8}\epsilon_{\alpha\beta\mu\nu}\Sigma^{\alpha\beta}\Sigma^{\mu\nu} = j_0 c, \tag{7}$$

where

$$j_0 = -\tfrac{1}{2}\xi\gamma^5 u, \tag{8}$$

with

$$\gamma^5 = -\gamma^1\gamma^2\gamma^3\gamma^4 \tag{9}$$

and

$$c = \tfrac{1}{2}\cdot\tfrac{1}{2}(\xi_a u^a + u^a \xi_a). \tag{10}$$

Equations (6) and (7) are, of course, manifestations of the well-known fact that the unitary irreducible representations of $\overline{\mathrm{SO}(3,1)}$ can be labeled by two numbers (j_0,c) (see Refs. 2–4). In Ref. 3, Dirac shows that the unitary irrep $(j_0,c) = (\frac{1}{2},0)$ provides a representation of SO(3,2). However, we emphasize that Dirac does not work on $L^2(D_4)$. In the Appendix we show that j_0 takes on half-integer values on $L^2(D_4)$ while c is not diagonal. Here, $|j_0|$ is the so-called "minimum spin" of the representation. This nomenclature arises from the fact that $\frac{1}{2}\Sigma^{jk}\Sigma_{jk}$, $j,k,\ldots = 1,2,3$ has eigenvalues $s(s+1)$, where s is constrained to satisfy $s \geqslant |j_0|$ (see Refs. 2 and 4).

Dirac spinor with components u^a, a = 1, 2, 3, 4; irrep, and DDξ, ...

- The Wigner D-function $D^j_{m_1,m_2}$ gives the matrix element of a rotation operator parametrized by Euler angles in a $2j+1$–dimensional unitary representation of a rotation group when parameters j, m_1, m_2 are physical, i.e. all integers or half-integers such that $-j \le m_1, m_2 \le j$.
- For unphysical parameters, WignerD is defined by an analytic continuation.

- The Wolfram Language uses phase conventions where $D^j_{m_1,m_2}(\psi, \theta, \phi) = \exp(i\, m_1\, \psi + i\, m_2\, \phi)\, D^j_{m_1,m_2}(0, \theta, 0)$.

WignerD[{j, m_1, m_2}, ψ, θ, ϕ]
gives the Wigner D-function $D^j_{m_1,m_2}(\psi, \theta, \phi)$.

In[134]:=
```
U1by4 = Array[uᵃ[#] &, 4]
```
Out[134]=
```
{uᵃ[1], uᵃ[2], uᵃ[3], uᵃ[4]}
```
In[135]:=
```
Protect[U1by4]
```
Out[135]=
```
{U1by4}
```
In[136]:=
```
r2 = U1by4.U1by4
```
Out[136]=
$u^a[1]^2 + u^a[2]^2 + u^a[3]^2 + u^a[4]^2$

In[137]:=
```
Head[U1by4]
```
Out[137]=
```
List
```
In[138]:=
```
Head[allMighty[##] & @@ U1by4]
```
Out[138]=
```
allMighty
```
In[139]:=
```
Head[Sin[x]]
```
Out[139]=
```
Sin
```
In[140]:=
```
Head[U1by4.U1by4]
```
Out[140]=
```
Plus
```
In[141]:=
```
processInput[input_] := Module[{f, vars},
  If[Head[input] === Function, f = input, vars = Variables[input];
   If[Length[vars] === 1, f = Function[{vars[[1]]}, Evaluate[input]],
    Return["Error: Expression must have only one variable."]]];
  (*Use'f' here*)f[10]]
```

$$\xi = -i\frac{\partial}{\partial u}, \tag{2}$$

In[142]:=
```
DDξ[f_] := -ⅈ * D[f, #] & /@ U1by4
```

In[143]:=
```
DDξ[U1by4]
```

Out[143]=
```
{{-ⅈ, 0, 0, 0}, {0, -ⅈ, 0, 0}, {0, 0, -ⅈ, 0}, {0, 0, 0, -ⅈ}}
```

In[144]:=
```
(*Symbolize[ S44αβ ]*)
```

In[145]:=
```
(*S44αβ=Table[FullSimplify[(γ[[h]].γ[[k]]-γ[[k]].γ[[h]])/4],{h,1,4},{k,1,4}];*)
```

In[146]:=
```
Dimensions[S44αβ]
```

Out[146]=
```
{4, 4, 4, 4}
```

In[147]:=
```
S44αβ[[1, 1]]
```

Out[147]=
```
{{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}}
```

$$\Sigma^{\alpha\beta} = \xi S^{\alpha\beta} u,$$

$$\xi = -i\frac{\partial}{\partial u}, \tag{2}$$

fixed:

In[148]:=
```
Clear[Σ^αβ];
Σ^αβ[f_](*/;Head[f]===Symbol*) := Table[Block[{Su, d},
   Su = f * S44αβ[[h, k]].U1by4;
   d = FullSimplify[Sum[-ⅈ * D[Su[[j]], U1by4[[j]]], {j, 1, Length[U1by4]}]];
   d], {h, 1, 4}, {k, 1, 4}]
```

In[150]:=
```
(*Clear[Σ^αβ];
  Σ^αβ[f_]:=Table[Block[{Su,d},
   Su=f*S44αβ[[h,k]].U1by4;
   d=FullSimplify[Sum[D[Su[[j]],U1by4[[j]]],{j,1,Length[U1by4]}]];
   d],{h,1,4},{k,1,4}]*)
```

$$\frac{1}{2}\Sigma^{jk}\Sigma_{jk} = \frac{1}{4}\partial_a\, u^a u^b\, \partial_b - \frac{1}{4}\partial_a\, u^b u^b\, \partial_a,$$

In[151]:=

```
(*Table[
 Sum[S44αβ⟦h,j⟧[[.S44αβ⟦j,k⟧,{j,1,4}],{h,1,3},{k,h+1,4}]//MatrixForm*)
```

In[152]:=

```
(sigma3 = {S44αβ⟦1, 2⟧, S44αβ⟦2, 3⟧, S44αβ⟦3, 1⟧}) // Column
MatrixForm[#] & /@ %⟦1⟧
```

Out[152]=

$$\left\{\left\{0, -\frac{1}{2}, 0, 0\right\}, \left\{\frac{1}{2}, 0, 0, 0\right\}, \left\{0, 0, 0, -\frac{1}{2}\right\}, \left\{0, 0, \frac{1}{2}, 0\right\}\right\}$$
$$\left\{\left\{0, 0, 0, \frac{1}{2}\right\}, \left\{0, 0, \frac{1}{2}, 0\right\}, \left\{0, -\frac{1}{2}, 0, 0\right\}, \left\{-\frac{1}{2}, 0, 0, 0\right\}\right\}$$
$$\left\{\left\{0, 0, -\frac{1}{2}, 0\right\}, \left\{0, 0, 0, \frac{1}{2}\right\}, \left\{\frac{1}{2}, 0, 0, 0\right\}, \left\{0, -\frac{1}{2}, 0, 0\right\}\right\}$$

Out[153]=

$$\left\{\begin{pmatrix} 0 & -\frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 \end{pmatrix}\right\}$$

In[154]:=

```
commutator[a_?SquareMatrixQ, b_?SquareMatrixQ] /;
  Dimensions[a] === Dimensions[b] := FullSimplify[a.b - b.a]
```

In[155]:=

```
commutator[sigma3⟦1⟧, sigma3⟦2⟧] === sigma3⟦3⟧
```

Out[155]=

```
True
```

In[156]:=

```
(*Clear[sigmaSquared];
  (sigmaSquared=Table[Block[{s2},
      s2=Table[S44αβ⟦h;;k1⟧.S44αβ⟦k1;;k⟧,{k1,1,4}];
      s2],{h,1,3},{k,h+1,4}])//MatrixForm*)
```

In[157]:=

```
(*Symbolize[ Σ^2 ]*)
```

$$\frac{1}{2}\Sigma^{jk}\Sigma_{jk} = \frac{1}{4}\,\partial_a\, u^a u^b\,\partial_b - \frac{1}{4}\partial_a\, u^b u^b\,\partial_a,$$

In[158]:=

```
Clear[Σ²];
Σ²[f_](*/;Head[f]===Symbol*) := Module[{u, n, sa, sb, d},
  u = U1by4; n = Length[u];
  sa = Sum[D[u[[a1]] * Sum[u[[b]] × D[f, u[[b]]], {b, 1, n}], u[[a1]]], {a1, 1, n}];
  sb = Sum[D[(Sum[u[[b]] × u[[b]], {b, 1, n}] × D[f, u[[a1]]]), u[[a1]]], {a1, 1, n}];
  (*d=sa-sb;*)
  d = FullSimplify[1/4 (sa - sb)];
  Return[d];]
```

In[160]:=

```
Σ²[#]/# & /@ U1by4
```

Out[160]=

$$\left\{\frac{3}{4}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right\}$$

In[161]:=

```
Σ²[U1by4.U1by4]
```

Out[161]=

0

$$\tfrac{1}{2}\Sigma^{jk}\Sigma_{jk} = \tfrac{1}{4}\,\partial_a\, u^a u^b\,\partial_b - \tfrac{1}{4}\partial_a\, u^b u^b\,\partial_a, \tag{14}$$

where $\partial_a = i\xi_a$. Equation (14) is equivalent to Dirac's Eq. (24). If we put

$$u^a u^a = r^2,\ u^a\,\partial_a = r\frac{\partial}{\partial r},$$

and

$$\partial_a\,\partial_a = \frac{1}{r^3}\frac{\partial}{\partial r}r^3\frac{\partial}{\partial r} + \frac{1}{r^2}\Delta_0,$$

where Δ_0 is the restriction of the Laplacian on $\mathbb{R}^4$ to the unit three-sphere, then $\frac{1}{2}\Sigma^{jk}\Sigma_{jk}$ reduces to

$$\tfrac{1}{2}\Sigma^{jk}\Sigma_{jk} = -\tfrac{1}{4}\Delta_0. \tag{15}$$

Here, Δ_0 has eigenvalues $-l(l+2)$, $l = 0,1,2,\ldots$ (see Ref. 5), so that $\frac{1}{2}\Sigma^{jk}\Sigma_{jk}$ has eigenvalues $s(s+1)$, $s = 0,\frac{1}{2},1,\ldots$.

In[162]:=

```
(*S44αβ*)
```

$j_0 = -\tfrac{1}{2}\xi\gamma^5 u,$

In[163]:=

```
(*Symbolize[ j0 ]*)
```

In[164]:=

```
Clear[j0];
j0[f_](*/;Head[f]===Symbol*) := Block[{Su, d},
  Su = f * γ[[5]].U1by4;
  d = FullSimplify[-1/2 Sum[-ⅈ * D[Su[[j]], U1by4[[j]]], {j, 1, Length[U1by4]}]];
  d]
```

Clear: j_0 is not a symbol or a valid string pattern.

$$c = \tfrac{1}{2}\cdot\tfrac{1}{2}(\xi_a u^a + u^a \xi_a).$$

In[166]:=

```
j0[U1by4.U1by4]
```

Out[166]=

0

In[167]:=

```
(*Symbolize[ cop ]*)
```

In[168]:=

```
Clear[cop];
cop[f_](*/;Head[f]===Symbol*) :=
 FullSimplify[1/4 ( Sum[ -ⅈ * D[f * U1by4[[j]], U1by4[[j]]], {j, 1, Length[U1by4]}] +
     Transpose[U1by4].Transpose[DDξ[f]])]
```

Clear: c_{op} is not a symbol or a valid string pattern.

In[170]:=

```
r2
```

Out[170]=

$u^a[1]^2 + u^a[2]^2 + u^a[3]^2 + u^a[4]^2$

In[171]:=

```
(*{alpha,beta,gamma}={-ArcTan[ua[2],ua[1]]+ArcTan[ua[4],ua[3]],
   2 ArcTan[ √(ua[3]^2+ua[4]^2) , √(ua[1]^2+ua[2]^2) ],
   ArcTan[ua[2],ua[1]]+ArcTan[ua[4],ua[3]]};*)
```

ψnj0ss3

aiWignerSmallDRefactored:

In[172]:=

```
(*Non-recursive,
single-expression Wigner Small-d.Uses the explicit summation
  formula (Wigner's form).Valid for arbitrary complex j,
m,mp,beta (though physically meaningful for spin).-No'If'
 statements used.-No recursion used.-Returns a symbolic
 Sum expression that can be differentiated.-Uses Gamma[n+1]
 instead of Factorial[n] for analytic continuity.*)
aiWignerSmallDRefactored[j_, mp_, m_, beta_] :=
 Module[{(*k*)}, (*Prefactor with Square Roots of Factorials*)
  Sqrt[Gamma[j + mp + 1] * Gamma[j - mp + 1] * Gamma[j + m + 1] * Gamma[j - m + 1]] *
   (*The Summation over k*)Sum[((*Alternating sign and phase
       (-1)^(k+mp-m)*)(-1)^(k + mp - m) * (*Trigonometric terms*)
       (Cos[beta / 2])^(2 * j + m - mp - 2 * k) * (Sin[beta / 2])^(mp - m + 2 * k)) /
     ((*Denominators:Gamma functions handle the truncation of the series*)
      (*1/(n!) becomes 1/Gamma[n+1] which is 0 for negative integers*)
      Gamma[j - mp - k + 1] * Gamma[j + m - k + 1] * Gamma[k + mp - m + 1] *
       Gamma[k + 1]), (*Sum from 0 to Infinity:Terms naturally vanish
        when k exceeds bounds due to 1/Gamma poles.GenerateConditions→
     False prevents Mathematica from wrapping the result in
      conditional'If' blocks about convergence.*)
    {k, 0, Infinity}, GenerateConditions → False]]
```

In[173]:=

```
Clear[ψnj0ss3];
ψnj0ss3[μ_, j0_, s_, s3_] :=
 Module[{n, rSquared, wf, alpha, beta, gamma, ret},
  {alpha, beta, gamma} = {-ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]],
    2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]],
    ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]]};
  n = s + μ;
  rSquared = u^a[1]^2 + u^a[2]^2 + u^a[3]^2 + u^a[4]^2; (*AKA, 1,  on  boundary  *)
  wf = 1/π * Sqrt[(2 * s + 1) * (n - s - 1)! / (n + s)!] * Exp[-rSquared / 2] *
    LaguerreL[n - s - 1, 2 * s + 1, rSquared] * Exp[-ⅈ * j0 * alpha] *
    aiWignerSmallDRefactored[s, -s3, j0, beta] * Exp[ⅈ * s3 * gamma];
  Return[wf];
 ]
```

In[175]:=

```
ParallelTable[Block[{a1, a2, a3, a4, a5, a6, a7, f}, f = ψnj0ss3[μ, j0, s, s3];
  a1 =
   FullSimplify[FullSimplify[Σ^2[f]/f] /. {u^a[1]^2 + u^a[2]^2 + u^a[3]^2 + u^a[4]^2 → r^2} /.
     {Sqrt[(u^a[1]^2 + u^a[2]^2)^2] → (u^a[1]^2 + u^a[2]^2),
      Sqrt[(u^a[3]^2 + u^a[4]^2)^2] → (u^a[3]^2 + u^a[4]^2)}];
  a2 =
   FullSimplify[FullSimplify[j0[f]/f] /. {u^a[1]^2 + u^a[2]^2 + u^a[3]^2 + u^a[4]^2 → r^2} /.
     {Sqrt[(u^a[1]^2 + u^a[2]^2)^2] → (u^a[1]^2 + u^a[2]^2),
      Sqrt[(u^a[3]^2 + u^a[4]^2)^2] → (u^a[3]^2 + u^a[4]^2)}];
  a3 = s * (s + 1);
  a4 = j0;
  a5 = {a1, a2};
  a6 = {a3, a4};
  a7 = {a5, a6};
  {{s, s3, j0, μ},
   If[a5 == a6, Style[a7, Red, Bold, 16], Style[a7, Blue, Italic, 12]]}],
 {s, 0, 2, 1/2}, {s3, -s, s}, {j0, -s, s}, {μ, 1, 3}]
```

Out[175]=

```
{{{{{{0, 0, 0, 1}, a7}, {{0, 0, 0, 2}, a7}, {{0, 0, 0, 3}, a7}}}},
 ... 3 ..., {{{{{2, -2, -2, 1}, If[{6, -(2 e^(r^2/2 + 4 i ... 1 ... + 1/2 (... 1 ...)) (u^a[1] + i u^a[2])^4)/((u^a[1]^2 + u^a[2]^2)^2)} == {6, -2},
      Style[a7, Red, Bold, 16], Style[a7, Blue, Italic, 12]]}, {{2, -2, -2, 2}, If[... 1 ...]},
    {{2, -2, -2, 3}, If[{6, -2 e^(r^2/2 + ... 1 ... + 1/2 ... 1 ...)} == {6, -2}, Style[a7, Red, Bold, 16],
      Style[a7, Blue, Italic, 12]]}}, ... 3 ..., {... 1 ...}}, ... 4 ...}}
```

Size in memory: 405.7 kB | + Show more | Show all | Iconize | Store full expression in notebook

howabout :

In[176]:=

```
howabout[s_, s1_, s2_] := (-1)^(s1-s2) Sqrt[(s - s1) ! (s + s1) ! (s - s2) ! (s + s2) !] *
  Exp[-I * s1 * -ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]]] *
  Exp[-I * s2 * ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]]] *
  If[s2 > s1, (If[0 == 2 s - s2 + s1, 1,
       Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s-s2+s1)]
      ((-1)^(s2-s1) HypergeometricPFQ[{-s + s2, -s - s1}, {1 + s2 - s1},
         -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
        Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(s2-s1)))/
    ((s - s2) ! (s2 - s1) ! (s + s1) !), (If[0 == 2 s + s2 - s1, 1,
       Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s+s2-s1)]
      (HypergeometricPFQ[{-s - s2, -s + s1}, {1 - s2 + s1},
         -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
        Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(-s2+s1)))/
    ((s + s2) ! (s - s1) ! (-s2 + s1) !)]
```

In[177]:=

```
Block[{j0 = -1 / 2, s = 3 / 2, s3 = 1 / 2, n = s + 1, f, r, r1, ret},
 f = Exp[-r2 / 2] LaguerreL[n - s - 1, 2 * s + 1, r2] *
   (*(WignerMatrix[j0,aWigner])*)howabout[s, s3, j0];
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]};
 ret]
```

Out[177]=

$$\left\{-\left(\left(2\, e^{i\,\mathrm{ArcTan}[u^a[2],u^a[1]]+2\,\mathrm{ArcTan}[u^a[4],u^a[3]]+\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2-u^a[3]^2-u^a[4]^2\right)}\sqrt{u^a[1]^2+u^a[2]^2}\left(u^a[3]^2+u^a[4]^2\right)\left(1-\frac{u^a[1]^2+u^a[2]^2}{2\left(u^a[3]^2+u^a[4]^2\right)}\right)\right)\Big/\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2}\right),\ \frac{11}{4}-\frac{u^a[1]^2+u^a[2]^2}{u^a[3]^2+u^a[4]^2},\ \frac{1}{2}+i\right\}$$

In[178]:=

```
Block[{j0 = 1 / 2, s = 3 / 2, s3 = -1 / 2, n = s + 1, f, r, r1, ret},
 f = Exp[-r2 / 2] LaguerreL[n - s - 1, 2 * s + 1, r2] *
   (*(WignerMatrix[j0,aWigner])*)howabout[s, s3, j0];
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]};
 ret]
```

Out[178]=

$$\left\{\left(2\, e^{-i\,\mathrm{ArcTan}[u^a[2],u^a[1]]+2\,\mathrm{ArcTan}[u^a[4],u^a[3]]+\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2-u^a[3]^2-u^a[4]^2\right)}\sqrt{u^a[1]^2+u^a[2]^2}\,\left(u^a[3]^2+u^a[4]^2\right)\left(1-\frac{u^a[1]^2+u^a[2]^2}{2\left(u^a[3]^2+u^a[4]^2\right)}\right)\right)\Big/\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2},\ \frac{11}{4}-\frac{u^a[1]^2+u^a[2]^2}{u^a[3]^2+u^a[4]^2},\ -\frac{1}{2}+i\right\}$$

In[179]:=

```
Block[{j0 = 3 / 2, s = j0, s3 = -1 / 2, n = s + 1, f, r, r1, ret},
 f = Exp[-r2 / 2] LaguerreL[n - s - 1, 2 * s + 1, r2] *
   (*(WignerMatrix[j0,aWigner])*)wignerDmatrixU[s, s3, j0];
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]};
 ret]
```

Out[179]=

$$\left\{e^{\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2-u^a[3]^2-u^a[4]^2\right)}\,\mathrm{wignerDmatrixU}\left[\frac{3}{2},-\frac{1}{2},\frac{3}{2}\right],0,0\right\}$$

In[180]:=

```
Block[
 {s = 3 / 2, s3 = -1 / 2, j0 = 1 / 2, n = s + 1, f, r, r1, ret, alpha, beta, gamma},
 {alpha, beta, gamma} = {-ArcTan[uᵃ[2], uᵃ[1]] + ArcTan[uᵃ[4], uᵃ[3]],
   2 ArcTan[Sqrt[uᵃ[3]^2 + uᵃ[4]^2], Sqrt[uᵃ[1]^2 + uᵃ[2]^2]],
   ArcTan[uᵃ[2], uᵃ[1]] + ArcTan[uᵃ[4], uᵃ[3]]};
 f = Exp[-r2 / 2] * LaguerreL[n - s - 1, 2 * s + 1, r2] * (wignerDmatrixU[s, s3, j0]) *
    Exp[-I * s3 * alpha] * Exp[-I * j0 * gamma] // FullSimplify;
 Print[f];
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]};
 ret]
```

$$e^{\frac{1}{2}\left(-2\, i\, \mathrm{ArcTan}[u^a[2],u^a[1]]-u^a[1]^2-u^a[2]^2-u^a[3]^2-u^a[4]^2\right)}\ \mathrm{wignerDmatrixU}\left[\frac{3}{2},-\frac{1}{2},\frac{1}{2}\right]$$

Out[180]=

$$\left\{e^{\frac{1}{2}\left(-2\, i\, \mathrm{ArcTan}[u^a[2],u^a[1]]-u^a[1]^2-u^a[2]^2-u^a[3]^2-u^a[4]^2\right)}\ \mathrm{wignerDmatrixU}\left[\frac{3}{2},-\frac{1}{2},\frac{1}{2}\right],\ \frac{u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2}{4\left(u^a[1]^2+u^a[2]^2\right)},\ -\frac{1}{2}\right\}$$

In[181]:=

```
Block[
 {s = 3 / 2, s3 = -1 / 2, j0 = 3 / 2, n = s + 1, f, r, r1, ret, alpha, beta, gamma},
 {alpha, beta, gamma} = {-ArcTan[uᵃ[2], uᵃ[1]] + ArcTan[uᵃ[4], uᵃ[3]],
   2 ArcTan[Sqrt[uᵃ[3]^2 + uᵃ[4]^2], Sqrt[uᵃ[1]^2 + uᵃ[2]^2]],
   ArcTan[uᵃ[2], uᵃ[1]] + ArcTan[uᵃ[4], uᵃ[3]]};
 (*r=WignerMatrix[1/2,aWigner];*)
 f = Exp[-r2 / 2] LaguerreL[n - s - 1, 2 * s + 1, r2] * (wignerDmatrixU[s, s3, j0]) *
    Exp[-I * j0 * alpha] * Exp[-I * s3 * gamma] // FullSimplify;
 Print[f];
 (*r=Σ²[f](*&/@Flatten[f]*)//Simplify;
 r1=;*)
 (*r1=Transpose[{r,Flatten[f]}];*)
 ret = {f, FullSimplify[Σ²[f] / f], FullSimplify[j₀[f] / f]};
 ret]
```

$$e^{\frac{1}{2} i \left(4 \text{ArcTan}[u^a[2],u^a[1]]-2 \text{ArcTan}[u^a[4],u^a[3]]+i \left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)\right)} \text{wignerDmatrixU}\left[\frac{3}{2}, -\frac{1}{2}, \frac{3}{2}\right]$$

Out[181]=

$$\left\{e^{\frac{1}{2} i \left(4 \text{ArcTan}[u^a[2],u^a[1]]-2 \text{ArcTan}[u^a[4],u^a[3]]+i \left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)\right)} \text{wignerDmatrixU}\left[\frac{3}{2}, -\frac{1}{2}, \frac{3}{2}\right], \frac{\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)\left(u^a[1]^2+u^a[2]^2+4\left(u^a[3]^2+u^a[4]^2\right)\right)}{4\left(u^a[1]^2+u^a[2]^2\right)\left(u^a[3]^2+u^a[4]^2\right)}, \frac{3}{2}\right\}$$

In[182]:=

```
Block[{ss = 3 / 2, s3 = -1 / 2, j0 = 1 / 2,
  n = ss + 1, f, h, s, s1, s2, r, r1, ret, alpha, beta, gamma},
 {alpha, beta, gamma} = {-ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]],
   2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]],
   ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]]};

 (*r=WignerMatrix[1/2,aWigner];*)
 h[s_, s1_, s2_] := (-1)^(s1-s2) Sqrt[(s - s1) ! (s + s1) ! (s - s2) ! (s + s2) !] *
   Exp[-I * s1 * alpha] * Exp[-I * s2 * gamma] * If[s2 > s1, (If[0 == 2 s - s2 + s1,
        1, Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s-s2+s1)]
       ((-1)^(s2-s1) HypergeometricPFQ[{-s + s2, -s - s1}, {1 + s2 - s1},
          -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
         Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(s2-s1)))/
     ((s - s2) ! (s2 - s1) ! (s + s1) !), (If[0 == 2 s + s2 - s1, 1,
        Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s+s2-s1)]
       (HypergeometricPFQ[{-s - s2, -s + s1}, {1 - s2 + s1},
          -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
         Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(-s2+s1)))/
     ((s + s2) ! (s - s1) ! (-s2 + s1) !)];
 f = Exp[-r2 / 2] LaguerreL[n - ss - 1, 2 * ss + 1, r2] * h[ss, s3, j0]
 (*(WignerMatrix[j0,aWigner])*);
 (*r=Σ^2[#]&/@Flatten[f]//Simplify;*)
 (*r1=Transpose[{r,Flatten[f]}];*)
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]}(*&/@r1*);
 ret]
```

Out[182]=

$$\left\{\left(2\, e^{\frac{1}{2} i \left(-\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)-\frac{1}{2} i \left(\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)+\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2-u^a[3\right.}\right.\right.$$

$$\left.\sqrt{u^a[1]^2+u^a[2]^2}\,\left(u^a[3]^2+u^a[4]^2\right)\left(1-\frac{u^a[1]^2+u^a[2]^2}{2\left(u^a[3]^2+u^a[4]^2\right)}\right)\right)\Big/$$

$$\left.\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2},\ \frac{15}{4},\ -\frac{1}{2}\right\}$$

In[183]:=

```
Block[{ss = 3 / 2, s3 = -1 / 2, j0 = -1 / 2,
  n = ss + 1, f, h, s, s1, s2, r, r1, ret, alpha, beta, gamma},
 {alpha, beta, gamma} = {-ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]],
   2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]],
   ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]]};

 (*r=WignerMatrix[1/2,aWigner];*)
 h[s_, s1_, s2_] := (-1)^(s1-s2) Sqrt[(s - s1) ! (s + s1) ! (s - s2) ! (s + s2) !] *
   Exp[-I * s1 * alpha] * Exp[-I * s2 * gamma] * If[s2 > s1, (If[0 == 2 s - s2 + s1,
       1, Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s-s2+s1)]
      ((-1)^(s2-s1) HypergeometricPFQ[{-s + s2, -s - s1}, {1 + s2 - s1},
         -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
        Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(s2-s1)))/
     ((s - s2) ! (s2 - s1) ! (s + s1) !), (If[0 == 2 s + s2 - s1, 1,
       Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s+s2-s1)]
      (HypergeometricPFQ[{-s - s2, -s + s1}, {1 - s2 + s1},
         -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
        Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(-s2+s1)))/
     ((s + s2) ! (s - s1) ! (-s2 + s1) !)];
 f = Exp[-r2 / 2] LaguerreL[n - ss - 1, 2 * ss + 1, r2] * h[ss, s3, j0]
 (*(WignerMatrix[j0,aWigner])*);
 (*r=Σ^2[#]&/@Flatten[f]//Simplify;*)
 (*r1=Transpose[{r,Flatten[f]}];*)
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]}(*&/@r1*);
 ret]
```

Out[183]=

$$\left\{\left(e^{\frac{1}{2} i \left(-\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)+\frac{1}{2} i \left(\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)+\frac{1}{2} \left(-u^a[1]^2-u^a[2]^2-u^a[3]^2-\left(u^a[3]^2+u^a[4]^2\right)^{3/2} \left(1-\frac{2 \left(u^a[1]^2+u^a[2]^2\right)}{u^a[3]^2+u^a[4]^2}\right)\right)\right.}\right)\Big/\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2}, \frac{15}{4}, -\frac{1}{2}\right\}$$

In[184]:=

```
Block[{ss = 3 / 2, s3 = 1 / 2, j0 = -1 / 2,
  n = ss + 1, f, h, s, s1, s2, r, r1, ret, alpha, beta, gamma},
 {alpha, beta, gamma} = {-ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]],
   2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]],
   ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]]};

 (*r=WignerMatrix[1/2,aWigner];*)
 h[s_, s1_, s2_] := (-1)^(s1-s2) Sqrt[(s - s1) ! (s + s1) ! (s - s2) ! (s + s2) !] *
   Exp[-I * s1 * alpha] * Exp[-I * s2 * gamma] * If[s2 > s1, (If[0 == 2 s - s2 + s1,
        1, Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s-s2+s1)]
       ((-1)^(s2-s1) HypergeometricPFQ[{-s + s2, -s - s1}, {1 + s2 - s1},
          -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
         Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(s2-s1)))/
     ((s - s2) ! (s2 - s1) ! (s + s1) !), (If[0 == 2 s + s2 - s1, 1,
        Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s+s2-s1)]
       (HypergeometricPFQ[{-s - s2, -s + s1}, {1 - s2 + s1},
          -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
         Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(-s2+s1)))/
     ((s + s2) ! (s - s1) ! (-s2 + s1) !)];
 f = Exp[-r2 / 2] LaguerreL[n - ss - 1, 2 * ss + 1, r2] * h[ss, s3, j0]
 (*(WignerMatrix[j0,aWigner])*);
 (*r=Σ^2[#]&/@Flatten[f]//Simplify;*)
 (*r1=Transpose[{r,Flatten[f]}];*)
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]}(*&/@r1*);
 ret]
```

Out[184]=

$$\left\{-\left(\left(2\, e^{-\frac{1}{2} i \left(-\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)+\frac{1}{2} i \left(\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)+\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2\right.}\right.\right.\right.$$

$$\sqrt{u^a[1]^2+u^a[2]^2}\left(u^a[3]^2+u^a[4]^2\right)\left(1-\frac{u^a[1]^2+u^a[2]^2}{2\left(u^a[3]^2+u^a[4]^2\right)}\right)\Bigg)\Bigg/$$

$$\left.\left.\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2}\right),\ \frac{15}{4},\ \frac{1}{2}\right\}$$

In[185]:=

```
Block[{ss = 3 / 2, s3 = -3 / 2, j0 = -1 / 2,
  n = ss + 1, f, h, s, s1, s2, r, r1, ret, alpha, beta, gamma},
 {alpha, beta, gamma} = {-ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]],
   2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]],
   ArcTan[u^a[2], u^a[1]] + ArcTan[u^a[4], u^a[3]]};

 (*r=WignerMatrix[1/2,aWigner];*)
 h[s_, s1_, s2_] := (-1)^(s1-s2) Sqrt[(s - s1) ! (s + s1) ! (s - s2) ! (s + s2) !] *
   Exp[-I * s1 * alpha] * Exp[-I * s2 * gamma] * If[s2 > s1, (If[0 == 2 s - s2 + s1,
       1, Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s-s2+s1)]
      ((-1)^(s2-s1) HypergeometricPFQ[{-s + s2, -s - s1}, {1 + s2 - s1},
         -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
        Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(s2-s1)))/
     ((s - s2) ! (s2 - s1) ! (s + s1) !), (If[0 == 2 s + s2 - s1, 1,
       Cos[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(2 s+s2-s1)]
      (HypergeometricPFQ[{-s - s2, -s + s1}, {1 - s2 + s1},
         -Tan[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^2]
        Sin[2/2 ArcTan[Sqrt[u^a[3]^2 + u^a[4]^2], Sqrt[u^a[1]^2 + u^a[2]^2]]]^(-s2+s1)))/
     ((s + s2) ! (s - s1) ! (-s2 + s1) !)];
 f = Exp[-r2 / 2] LaguerreL[n - ss - 1, 2 * ss + 1, r2] * h[ss, s3, j0]
 (*(WignerMatrix[j0,aWigner])*);
 (*r=Σ^2[#]&/@Flatten[f]//Simplify;*)
 (*r1=Transpose[{r,Flatten[f]}];*)
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]}(*&/@r1*);
 ret]
```

Out[185]=

$$\left\{\left(\sqrt{3}\, e^{\frac{3}{2} i \left(-\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)+\frac{1}{2} i \left(\text{ArcTan}[u^a[2],u^a[1]]+\text{ArcTan}[u^a[4],u^a[3]]\right)+\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2-u^a[3\right.}\right.\right.$$

$$\left.\left.\sqrt{u^a[1]^2+u^a[2]^2}\left(u^a[3]^2+u^a[4]^2\right)\right)\Big/\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2},$$

$$\left.\frac{15}{4}, -\frac{3}{2}\right\}$$

In[186]:=

```
Clear[wignerDmatrixU];
wignerDmatrixU[j_, m1_, m2_] := (-1)^(-m1+m2)
  e^(-i m2 (-ArcTan[u^a[2],u^a[1]]+ArcTan[u^a[4],u^a[3]])-i m1 (ArcTan[u^a[2],u^a[1]]+ArcTan[u^a[4],u^a[3]]))
  Sqrt[(j - m1) ! (j + m1) ! (j - m2) ! (j + m2) !] Sum[(-1)^s
      If[2 j + m1 - m2 - 2 s == 0, 1, (Sqrt[u^a[3]^2 + u^a[4]^2]/Sqrt[u^a[1]^2 + u^a[2]^2 + u^a[3]^2 + u^a[4]^2])^(2 j+m1-m2-2 s)] ×
      If[-m1 + m2 + 2 s == 0, 1, (Sqrt[u^a[1]^2 + u^a[2]^2]/Sqrt[u^a[1]^2 + u^a[2]^2 + u^a[3]^2 + u^a[4]^2])^(-m1+m2+2 s)], {s, Max[0,m1-m2], Min[j+m1,j-m2]}] /
    ((j + m1 - s) ! (j - m2 - s) ! s ! (-m1 + m2 + s) !)
```

In[187]:=

```
Block[{ss = 3 / 2, s3 = -3 / 2, j0 = -1 / 2, n = ss + 1, f, h, s, s1, s2, r, r1, ret},

 f = Exp[-r2 / 2] LaguerreL[n - ss - 1, 2 * ss + 1, r2] *
    wignerDmatrixU[ss, s3, j0] // FullSimplify;
 ret = {f, FullSimplify[Σ^2[f]/f], FullSimplify[j0[f]/f]};
 ret]
```

Out[187]=

$$\left\{\frac{\sqrt{3}\, e^{\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2-u^a[3]^2-u^a[4]^2\right)}\left(i\, u^a[1]+u^a[2]\right)\left(u^a[3]-i\, u^a[4]\right)^2}{\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2}}, \frac{15}{4}, -\frac{1}{2}\right\}$$

In[188]:=

```
Block[{ss = 3 / 2, s3 = -3 / 2, j0 = -3 / 2, n = ss + 1, f, h, s, s1, s2, r, r1, ret},

 f = Exp[-r2 / 2] LaguerreL[n - ss - 1, 2 * ss + 1, r2] *
    wignerDmatrixU[ss, s3, j0] // FullSimplify;
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]};
 ret]
```

Out[188]=

$$\left\{-\frac{i\, e^{\frac{1}{2}\left(-u^a[1]^2-u^a[2]^2-u^a[3]^2-u^a[4]^2\right)}\left(u^a[3]-i\, u^a[4]\right)^3}{\left(u^a[1]^2+u^a[2]^2+u^a[3]^2+u^a[4]^2\right)^{3/2}},\ \frac{15}{4},\ -\frac{3}{2}\right\}$$

In[ ]:=

```
Block[{ss = 3 / 2, s3 = -3 / 2, j0(*=-3/2*), n = ss + 1, f, h, s, s1, s2, r, r1, ret},

 f = Exp[-r2 / 2] LaguerreL[n - ss - 1, 2 * ss + 1, r2] *
    wignerDmatrixU[ss, s3, j0] // FullSimplify;
 ret = {f, FullSimplify[Σ^2[f] / f], FullSimplify[j0[f] / f]};
 ret]
```

Out[ ]=

$Aborted